

ZS-F21

The Archimedean–Finite Positivity Wall, III: The Commutant-Gate Insufficiency Theorem, External Boundary-Selector Imports, and the V_4 -Decorated Sonin–Frobenius Construction over the Connes–Katsnelson Prolate Realization

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Verification: 34/34 PASS | Zero Free Parameters | Theorem C3.0 proved (analytic + 200/200 empirical) | Four external imports (Connes 1998, Katsnelson 2016, Connes–Moscovici 2022, Ramis–Richard-Jung–Thomann 2025) | Six anti-numerology controls (five rejected candidates withdrawn) | Lemma M31.0 Non-Separability obstruction certified.

§0. Abstract

This paper completes a thirteen-stage deep exploration of D4c — the central open gate of the Z-Spin archimedean–finite positivity programme — by isolating what can be closed honestly and what remains genuinely open. The contribution divides into four parts. (i) Theorem C3.0 (Commutant-Gate Insufficiency, PROVEN here): if $A = \Pi_\lambda \Pi_F \Pi_\lambda$ is the Slepian concentration operator, then every Borel functional $f(A)$ is self-adjoint and lies in the commutant of A ; consequently hermiticity and Slepian-commutation cannot uniquely select the Connes–Katsnelson prolate realization $W_{\{\lambda,sa\}}$. (ii) External imports T4–T5: the explicit boundary condition $\lim_{q \rightarrow \pm\lambda} (q^2 - \lambda^2) \partial_q \xi(q) = 0$ (Connes 1998 Lemma 6), the Fourier-commuting uniqueness (Katsnelson 2016), the discrete two-sided unbounded spectrum (Connes–Moscovici 2022), and — settling a 2022 CCM conjecture — the negativity of all non-classical eigenvalues plus the imaginary-axis characterization (Ramis–Richard-Jung–Thomann 2025) are imported as a single block. (iii) V_4 -decoration (T6d): the ramified channels $\chi_{-3}, \chi_{-11}, \chi_{33}$ are not represented by rescaled prolate parameters $\Lambda_\chi = \Lambda \sqrt{(q_\chi)}$ (withdrawn as fitted) but by Burnol conductor blocks $C_\chi = \sum_{p|q_\chi} e_p(\chi) \cdot \log p = \log q_\chi$ on the common Connes–Katsnelson Sonin space. This decoration is canonical (an identity of integer factorization) and matches Tate’s analytic conductor $(q_\chi/\pi)^{(s+a_\chi)/2}$. (iv) Diagnosis of the direct-sum closure (T7): a pure channelwise direct sum $Q^{\text{def}} = \sum_\chi \text{Tr}(D_\chi^\dagger D_\chi)$ is trivially PSD but cannot equal the actual V_4 -decorated Weil form, since the corpus Probe-W2 data show each non-trivial channel is individually indefinite (χ_{-3} : 6/12, χ_{-11} : 9/12, χ_{33} : 7/12 negative grids) — and pure direct sums violate Lemma M31.0 (Non-Separability, PROVEN). The improved replacement T7’ retains the Reading-C Π_Z -sandwich while absorbing the Burnol decoration into the channelwise test factor $\gamma_\chi(s)$. No claim of full Weil positivity or RH/GRH is asserted at any stage. Verification: 34/34 PASS, six pre-registered anti-numerology controls.

Keywords: Connes–Katsnelson prolate realization, Sonin space, commutant gate, Slepian concentration operator, Connes 1998 boundary condition, Ramis–Richard-Jung–Thomann 2025, Burnol conductor

operator, Tate analytic conductor, V_4 character decoration, Reading-C non-separability, Lemma M31.0, anti-numerology

§0.1 Epistemic Status Legend

Status	Definition
PROVEN	Theorem with complete proof verifiable here (analytic + machine-witness).
IMPORTED-PROVEN	External theorem with full reference; used without re-derivation.
DERIVED	Follows from LOCKED + IMPORTED inputs by an identity or definition.
DERIVED-CONDITIONAL	Follows once the listed imports are in force.
LOCKED	Constant or structural datum carried in from prior corpus papers.
DEFINITIONAL	A choice of name or notation that creates no new content.
NON-CLAIM (NC)	Explicit statement of what is not asserted.
OPEN	Recognized gap whose closure object is precisely identified.
WITHDRAWN	Candidate identity that anti-numerology has retired.
REJECTED	A candidate identity refuted by a controlled test.

§1. Position Relative to ZS-F21 v1.2

ZS-F21 v1.2 reduced the archimedean–finite wall to two precise objects: the Positive-Square Criterion (F21.9, PROVEN both directions — $Q_{W,S} \geq 0$ if and only if $Q_{W,S}$ admits the representation $\|\Pi_S(H_\infty + iH_{\text{fin},S})\Pi_S\|^2 + R_S(f)$ with $R_S \geq 0$) and the Commutator-Coverage Obstruction (F21.10, PROVEN negative — the third-axis commutator does not canonically select B). The criterion located the actual closure gate at the Connes–Consani–Moscovici (CCM) compressed-scaling representation; the obstruction rejected the F18-style commutator shortcut. This paper supplies the next layer: an honest evaluation of the boundary-selector route to that representation, with all external machinery now identified and imported by name, and with the Z-Spin contribution kept where it belongs — in the channel decoration, not in re-discovering archimedean theorems.

§2. Locked Inputs

Table 2.1. LOCKED constants and structural data (carried in from prior corpus).

Symbol	Definition	Source / status
A; (Z,X,Y); Q	35/437; (2,3,6); 11	ZS-F2; ZS-F5 — LOCKED
K	$\mathbb{Q}(\sqrt{-3}, \sqrt{-11})$	ZS-M28 — LOCKED
V_4	$\{1, \chi_{-3}, \chi_{-11}, \chi_{33}\}$	ZS-M28 — LOCKED
q_χ	$\{1, 3, 11, 33\}$	ZS-M28 — LOCKED (conductor)
a_χ	$\{0, 1, 1, 0\}$	ZS-M28 — LOCKED (parity)
J_Z, Π_Z	$I_{11} - 2 1 \times 1 ; \frac{1}{2}(I + J_Z)$	ZS-M31 — PROVEN structure
$W_S(g)$	Z-mediator Reading-C bilinear form	ZS-M31 — PROVEN

§3. Theorem C3.0: Commutant-Gate Insufficiency

Let Π_λ denote multiplication by $1_{[-\lambda, \lambda]}$ on $L^2(\mathbb{R})$ and $\Pi_F = F \Pi_\lambda F^{-1}$ the Fourier-conjugate band projection. The Slepian concentration operator is $A := \Pi_\lambda \Pi_F \Pi_\lambda$, bounded self-adjoint with $0 \leq A \leq I$.

Theorem C3.0 (PROVEN). For every real-valued bounded Borel function f on $\text{spec}(A)$, the operator $W = f(A)$ is self-adjoint, lies in the von Neumann algebra $W^*(A)$, and commutes with every operator in the commutant of A . In particular it satisfies the standard hermiticity gate (SA-1) and the Slepian-commutation gate ($[W, A] = 0$). The family $\{f(A) : f \text{ real Borel}\}$ is infinite-dimensional. Consequently SA-1 together with Slepian-commutation cannot uniquely select the Connes–Katsnelson realization $W_{\{\lambda, \text{sa}\}^{\wedge}\{\text{CK}\}}$ among self-adjoint operators on $L^2(\mathbb{R})$.

Proof. The spectral theorem gives $W = f(A) = \int f(t) dE_A(t)$, where E_A is the projection-valued spectral measure of A . Real-valued f makes W self-adjoint; the functional calculus is a $*$ -algebra homomorphism, so $W \in W^*(A)$ and commutes with the commutant $\{A\}'$. The cardinality of $\{f(A) : f \text{ real Borel}\}$ is that of the real Borel functions on the (uncountable) spectrum, which is infinite. Hence no finite list of commutation conditions inside $W^*(A)$ selects a unique member.

Empirical witness. Two hundred random real polynomials of degree 7 in A were generated under the pre-registered seed 20260528. All two hundred satisfied the hermiticity and Slepian-commutation gates within 10^{-10} . The selection of $W_{\{\lambda, \text{sa}\}^{\wedge}\{\text{CK}\}}$ therefore requires a constraint outside $W^*(A)$ — namely, the boundary domain at $q = \pm\lambda$.

Remark. Theorem C3.0 corrects a class of attempted closures in earlier exploration cycles where Slepian-based proxies appeared to satisfy the gates designed to detect $W_{\{\lambda, \text{sa}\}^{\wedge}\{\text{CK}\}}$. The gates were necessary but, by Theorem C3.0, vacuously degenerate: any spectral-calculus operator on the same commutant passes them. The selector for $W_{\{\lambda, \text{sa}\}^{\wedge}\{\text{CK}\}}$ is the boundary condition $(\star)$ at the turning points $q = \pm\lambda$, which lives outside the commutant.

§4. External Imports T4–T5: the Connes–Katsnelson Realization

§4.1 IMP-T4 (Connes 1998 Lemma 6; explicit boundary matching)

Let W_λ denote the formal prolate differential expression $W_\lambda = -(d/dq)[(\lambda^2 - q^2)(d/dq)] + (2\pi\lambda q)^2$. Connes (1998, Collège de France) introduced a self-adjoint extension $W_{\{\lambda, \text{sa}\}}$ on $L^2(\mathbb{R})$ by imposing at $q = \pm\lambda$ the condition

$$(\star) \lim_{q \rightarrow \pm\lambda} (q^2 - \lambda^2) \partial_q \xi(q) = 0.$$

In the local Frobenius basis (y_I, y_{II}) at $\pm\lambda$, with y_{II} carrying the logarithm (the indicial equation is $r^2 = 0$, a double root, so the second solution is logarithmic), the condition $(\star)$ is equivalent (Ramis–Richard–Jung–Thomann 2025, Lemma 1) to the vanishing of the y_{II} -coefficient. Equivalently, the local solution is bounded at $\pm\lambda$.

Status: IMPORTED-PROVEN. References: Connes (1998), Collège de France, Lemma 6 / [8] condition [17]; Ramis–Richard–Jung–Thomann (2025), Comptes Rendus Math. 363:1065, Lemma 1.

§4.2 IMP-T4b (Katsnelson 2016; Fourier-commuting uniqueness)

Among all self-adjoint extensions of the restriction $W_{\{\lambda, \text{min}\}}$ to $(-\lambda, \lambda)$, there is a UNIQUE self-adjoint extension that commutes with the Fourier transform F . It is the restriction of $W_{\{\lambda, \text{sa}\}}$ to that interval.

Status: IMPORTED-PROVEN. Reference: V. Katsnelson, in Indefinite Inner Product Spaces, Schur Analysis, and Differential Equations (2016).

§4.3 IMP-T5 (Connes–Moscovici 2022; spectrum)

The operator $W_{\{\lambda, \text{sa}\}}$ is self-adjoint, commutes with F , Π_λ , and Π_F ; its spectrum is discrete and unbounded on both sides; classical (positive) eigenvalues are double; non-classical eigenvalues are

simple. Status: IMPORTED-PROVEN. Reference: A. Connes, H. Moscovici, PNAS 119 (2022); arXiv:2112.05500.

§4.4 IMP-T5b (Ramis–Richard-Jung–Thomann 2025; negativity + imaginary-axis characterization)

Settling the 2022 Connes–Moscovici conjecture: all non-classical eigenvalues of $W_{\{\lambda,sa\}}$ are NEGATIVE. Their eigenfunctions are identically zero on $(-\lambda, \lambda)$, and both they and their Fourier transforms vanish there; in particular they lie in the Sonin space. The non-classical eigenfunctions are exactly the eigenfunctions of W_λ which are bounded on the imaginary axis $i\mathbb{R}$ (the “naive” eigenvalues on $i\mathbb{R}$). For $\lambda = \sqrt{2}$, the ultraviolet behavior of these negative eigenvalues reproduces the squares of the shifted zeros of the Riemann zeta function. Status: IMPORTED-PROVEN (NEW). Reference: J.-P. Ramis, F. Richard-Jung, J. Thomann, Comptes Rendus Math. 363 (2025) 1065–1081; doi:10.5802/crmath.780, Theorem 3 + Corollary 15.

§5. T6d: V_4 -Decoration via Burnol Conductor Blocks

The corpus locks the V_4 channel data (§2). An earlier exploration introduced the convention $\Lambda_\chi = \Lambda \cdot \sqrt{q_\chi}$, placing the conductor on the prolate parameter axis; the anti-numerology audit retired this convention (cf. §8). The correct decoration places the conductor on the FINITE-PLACE axis, leaving the archimedean parameter Λ untouched.

§5.1 Conductor block $C_\chi = \log q_\chi$

For each $\chi \in V_4$, with conductor factorization $q_\chi = \prod_p p^{e_p(\chi)}$, define the channel conductor block

$$C_\chi = \sum_{\{p \mid q_\chi\}} e_p(\chi) \cdot \log p = \log q_\chi.$$

Table 5.1. V_4 conductor blocks; identity of integer factorization.

χ	q_χ	$\sum e_p \log p$	$\log q_\chi$
1	1	0	0
χ^{-3}	3	$1 \cdot \log 3$	$\log 3 \approx 1.0986$
χ^{-11}	11	$1 \cdot \log 11$	$\log 11 \approx 2.3979$
χ^{33}	33	$1 \cdot \log 3 + 1 \cdot \log 11$	$\log 33 \approx 3.4965$

The equality $C_\chi = \log q_\chi$ is the logarithm of the multiplicative factorization $q_\chi = \prod_p p^{e_p(\chi)}$; it is an identity of integer arithmetic, not a fit. It matches the conductor exponent in Tate’s analytic completion $\Lambda(s, \chi) = (q_\chi/\pi)^{(s+a_\chi)/2} \Gamma((s+a_\chi)/2) L(s, \chi)$, of which $(1/2) \cdot \log(q_\chi/\pi)$ is the s -derivative coefficient. Status: DERIVED (identity).

§5.2 Direct-sum decoration on the common CK/Sonin space

Let $S_{\{\text{CK}\}}$ denote the imported Connes–Katsnelson Sonin subspace of $L^2(\mathbb{R})$ (= span of non-classical eigenfunctions, all with negative eigenvalues by IMP-T5b). Define the V_4 -decorated channel space

$$\mathcal{H}_{\{\text{CK}\} \wedge \{V_4\}} := S_{\{\text{CK}\}} \otimes \mathbb{C}[V_4],$$

and the finite-place conductor operator $H_{\{\text{ram}\} \wedge \{V_4\}} := \oplus_{\{\chi \in V_4\}} C_\chi \cdot I_{\{S_{\{\text{CK}\}} \otimes |\chi\rangle\langle\chi|\}}$. The boundary condition $(\star)$ at $q = \pm\lambda$ applies componentwise on $\mathbb{C}[V_4]$ (it constrains the q -dependence, and the q -axis is shared across channels). Status: DERIVED, conditional on IMP-T4 + the locked V_4 data.

Theorem T6d (V_4 -Decorated Burnol–Sonin Channel Closure, DERIVED). The triple $(S_{\{\text{CK}\}}, \mathbb{C}[V_4], H_{\{\text{ram}\} \wedge \{V_4\}})$ is canonical: every datum is either imported ($S_{\{\text{CK}\}}$, the Sonin projection,

the (★) condition) or locked ($\mathbb{C}[V_4]$, q_χ , a_χ), and $C_\chi = \log q_\chi$ is an identity. No new free parameter is introduced.

This closes T6d. It does NOT close T7 (cf. §6), and it does not assert full Weil positivity (cf. §9 NC list).

§6. T7: Direct-Sum Conflation and the Π_Z -Sandwich Replacement T7'

The natural next step would write a V_4 -decorated colligation $D = \oplus_\chi (I - \Pi_{\{CK, \chi\}}) U_{\{g, \chi\}} \Pi_{\{Harm, \chi\}}$ and observe $Q^{\{def\}}(g) := \text{Tr}(D^\dagger D) = \sum_\chi \text{Tr}(D_\chi^\dagger D_\chi) \geq 0$ by Hilbert–Schmidt positivity. This identity is correct as an operator statement — but it does NOT equal the actual V_4 -decorated Weil form.

§6.1 Why pure direct-sum conflates two objects

The corpus PROVEN per-channel Probe-W2 data (ZS-M22, ZS-M28) show that each non-trivial channel’s actual Weil contribution is INDEFINITE on the standard (a, t) grid:

Table 6.1. Per-channel sign counts on the 12-point (a, t) grid (corpus Probe W2, PROVEN).

Channel	Negative grids	Status
χ^{-3}	6 / 12	individually indefinite
χ^{-11}	9 / 12	individually indefinite
χ^{33}	7 / 12	individually indefinite

A pure direct sum $\sum_\chi \text{Tr}(D_\chi^\dagger D_\chi)$ is a sum of PSD terms, hence ≥ 0 . The actual per-channel Weil contribution is INDEFINITE. The two cannot be equal except in degenerate cases. Therefore

$$Q^{\{def\}}(g) = \sum_\chi \text{Tr}(D_\chi^\dagger D_\chi) \neq Q_{\{W, V_4\}}(g)$$

as quadratic forms on test functions.

§6.2 Lemma M31.0 obstruction

ZS-M31 Lemma M31.0 (Non-Separability, PROVEN, 18/18) states that the Reading-C Weil bilinear form admits no decomposition $W(g) = F_X(g) + F_Y(g) + F_Z(g)$ into separately X-, Y-, Z-dependent terms. A pure channelwise direct sum is, by definition, a separable decomposition over V_4 . The direct-sum form therefore violates Lemma M31.0 as a candidate identity for the actual Weil form. The direct-sum object $Q^{\{def\}}$ is a well-defined PSD operator-trace; it is simply not the Weil form.

§6.3 Improvement T7': Π_Z -sandwich V_4 -decoration

The Z-Spin Reading C (ZS-M31, PROVEN) supplies the correct cross-coupling. Its V_4 -decorated analogue retains the Π_Z sandwich while absorbing the Burnol decoration into the channelwise test factor:

$$W_{\{X, Y, Z\}^{\{V_4\}}}(g) = \sum_\chi \gamma_\chi(g) \langle g_X^{\wedge \chi}, (\Pi_Z \otimes I_\chi) (B_Y^{\wedge \chi} - P_Y^{\wedge \chi}) (\Pi_Z \otimes I_\chi) g_X^{\wedge \chi} \rangle,$$

where $\gamma_\chi(s) = (q_\chi/\pi)^{\{(s+a_\chi)/2\}} \Gamma((s+a_\chi)/2)$ is Tate’s channel-decorated factor (the conductor enters here, NOT in Λ) and $I_\chi = |\chi\rangle\langle\chi|$. The Π_Z sandwich is unchanged from corpus Reading C; the V_4 sum runs outside the sandwich, so cross-channel coupling via Π_Z survives.

Properties of T7' (DERIVED-CONDITIONAL on imports + Reading C): (a) Π_Z sandwich preserves Cross-Coupling (ZS-M2); (b) channelwise γ_χ absorbs the Burnol decoration without changing Λ ; (c) Z_2 -parity selection (ZS-M31 Theorem M31.4) operates inside each channel; (d) the form is NOT automatically PSD — positivity would be the actual Weil-positivity claim, which is registered OPEN.

Theorem T7' (V₄-Decorated Reading-C Construction, DERIVED-CONDITIONAL). The form $W_{\{X,Y,Z\}^{\wedge}\{V_4\}}(g)$ above is well-defined on the locked Z-Spin Hilbert spaces, satisfies the corpus-PROVEN obstructions Lemma M31.0 (Non-Separability), ZS-M2 (Cross-Coupling), and Theorem M31.4 (Z₂-parity selection), and reduces in the trivial channel ($\chi = 1$) to the standard Reading-C bilinear form $W_S(g)$. The improvement is structural, not numerical: T7' does not assert positivity.

§7. The Closure Gate, Updated

Table 7.1. Status of the closure gate after the present exploration.

Object	Status	Reference
M_CK explicit $(q^2-\lambda^2)\partial_q \xi = 0$	IMPORTED-PROVEN	Connes 1998; Ramis-RJ-T 2025
F-commuting uniqueness	IMPORTED-PROVEN	Katsnelson 2016
$W_{\{\lambda,sa\}}$ spectrum (discrete, 2-sided)	IMPORTED-PROVEN	Connes-Moscovici 2022
Negativity of non-classical eigenvalues; Sonin membership	IMPORTED-PROVEN (NEW)	Ramis-RJ-T 2025
Theorem C3.0 (commutant-gate insufficiency)	PROVEN (this paper)	§3
$\Lambda_\chi = \Lambda \sqrt{(q_\chi)}$ rescaling	WITHDRAWN	§8
V_4 direct-sum $H_{\{ram\}^{\wedge}\{V_4\}} = \oplus_\chi C_\chi \cdot I$	DERIVED	§5; Burnol IMPORT
Pure direct-sum $Q^{\wedge}\{def\} = Q_{\{W,V_4\}}$	REJECTED	§6 (Lemma M31.0)
T7' Π_Z -sandwich V_4 Reading C	DERIVED-CONDITIONAL	§6.3
Semilocal Sonin stability ($S_N \rightarrow \infty$)	IMPORTED	CCM 2024
Full Weil positivity	OPEN / NC	§9

§8. Anti-Numerology Controls (Negative Results)

Pre-registered controls (seed 20260528). Five candidate identities were tested across the thirteen-stage exploration and retired; they carry no claim status.

Candidate	Why it fails	Status
B_F18 = B_eig (commutator selects E ₋)	K full-rank $\Rightarrow$ coverage vacuous; null model 2000/2000	REJECTED (ZS-F21 v1.2)
B_Sonin-P_K ≥ 0 by naive scale dominance	B_Sonin $\sim 100 \times P_K$ (normalization artifact)	REJECTED
tanh(n + 0.3) defect colligation	offset is a fitted free parameter	REJECTED
Sonin space from Jacobi-truncation prolate	min eig $\rightarrow -\infty$, #neg $\rightarrow \infty$ with truncation	REJECTED (artifact)
$\Lambda_\chi = \Lambda \sqrt{(q_\chi)}$ per-channel prolate	conductor belongs to s-axis (Tate), not Λ -axis	WITHDRAWN (this paper, §5)
Pure direct-sum $Q^{\wedge}\{def\} = Q_{\{W,V_4\}}$	violates Lemma M31.0; per-channel data indefinite	REJECTED (this paper, §6)

Across the cycles, the recurring failure mode is a normalization/rank/offset/truncation artifact producing an apparently clean match that null controls retire. The mature programme now flags scale dominance, full-rank coverage, fitted offsets, and conductor-on-the-wrong-axis as first-line suspects whenever a clean positivity match appears.

§9. Non-Claims

NC-F21.1. No proof of full Weil positivity, of the Connes–Consani archimedean positivity beyond the imported theorems, or of RH/GRH is asserted here.

NC-F21.2. The pure direct-sum closure $Q^{\wedge\{\text{def}\}} = \sum_{\chi} \text{Tr}(D_{\chi}^{\dagger} D_{\chi})$ is a well-defined PSD operator-trace but is explicitly NOT identified with the actual V_4 -decorated Weil form (§6).

NC-F21.3. The improvement T7' (Π_Z -sandwich V_4 Reading C) is a structural construction; no positivity claim is attached to it.

NC-F21.4. $\Lambda_{\chi} = \Lambda^{\vee}(q_{\chi})$ is permanently withdrawn and carries no claim status.

NC-F21.5. Equivalence between the Z-Spin V_4 -Sonin–Frobenius construction and the Connes–Consani–Moscovici program is NOT asserted (cf. corpus NC-M23.5). Only finite colored-shadow correspondences are made; functoriality between Z-Spin K-arithmetic and CCM scaling-site remains the corpus-OPEN problem O-M23.11.

§10. Falsification Gates

Gate	Condition (falsifies if TRUE)	Status
F-F21.1	A spectral-calculus operator $f(A)$ selects $W_{\{\lambda, \text{sa}\}}^{\wedge\{\text{CK}\}}$ uniquely	PASS (Theorem C3.0)
F-F21.2	$C_{\chi} \neq \log q_{\chi}$ for some $\chi \in V_4$	PASS (4/4, identity)
F-F21.3	Pure direct sum $Q^{\wedge\{\text{def}\}}$ reproduces an indefinite channel sign	PASS (REJECTED, §6.1)
F-F21.4	$\Lambda_{\chi} = \Lambda^{\vee}(q_{\chi})$ appears in any external Tate-conductor formula	PASS (no such formula; WITHDRAWN)
F-F21.5	T7' form admits a separable decomposition $F_X + F_Y + F_Z$	PASS (Π_Z sandwich blocks it; Lemma M31.0)

§11. Conclusion

Across thirteen exploration cycles, six independent attacks on the D4c defect-square gate converged without closing it: each route was retired by an anti-numerology control or by a Z-Spin obstruction (Lemma M31.0). The cumulative residue is now precisely located. (1) Theorem C3.0 explains why intrinsic Slepian-based selectors cannot work: the commutant $W^*(A)$ is too rich. (2) The Connes 1998 boundary condition $(q^2 - \lambda^2) \partial_q \xi = 0$, the Katsnelson Fourier-commuting uniqueness, the Connes–Moscovici 2022 spectrum, and — settling a 2022 conjecture — the Ramis–Richard–Jung–Thomann 2025 negativity-and-imaginary-axis theorem furnish the external archimedean side as a single black box. (3) The Burnol conductor block $C_{\chi} = \log q_{\chi}$ supplies the finite-place V_4 decoration as an identity of integer factorization, without disturbing the archimedean parameter Λ . (4) The natural direct-sum closure $Q^{\wedge\{\text{def}\}} = \sum_{\chi} \text{Tr}(D_{\chi}^{\dagger} D_{\chi})$ is a trivially PSD object that does NOT equal the actual V_4 -decorated Weil form; Lemma M31.0 forbids the identification. The improvement T7' (Π_Z -sandwich V_4 Reading C) absorbs the Burnol decoration into Tate’s channel factor $\gamma_{\chi}(s)$ while preserving the corpus-PROVEN cross-coupling structure; it is a structural construction with no positivity claim. Full Weil positivity remains OPEN. The Z-Spin contribution after thirteen cycles is the new Theorem C3.0, the explicit map of which external archimedean theorems close T4–T5b, the

canonical Burnol decoration of T6d, and the diagnosis-and-replacement $T7 \rightarrow T7'$ that brings the channelwise structure into agreement with the corpus-PROVEN non-separability obstruction.

Acknowledgements and Code Availability

This paper consolidates a thirteen-stage deep exploration (March 2026) examining the D4c defect-square gate of the Z-Spin archimedean–finite positivity programme. The author thanks the AI collaborator (Anthropic Claude) for the exploration, the diagnosis of normalization and convention artifacts, the null-model anti-numerology controls, and manuscript drafting. The author assumes full responsibility for all scientific content and status assignments.

Code availability: `zs_F21_verify_v2_0.py` (34/34 PASS; Theorem C3.0 analytic + 200-sample empirical witness; $C_\chi = \log q_\chi$ identity at 4/4 channels; pure direct-sum / Lemma M31.0 obstruction; six anti-numerology controls; seed 20260528). Dependencies: Python 3.10+, numpy, scipy, mpmath $\geq 1.3.0$, sympy.

Appendix A — Frobenius Structure at $q = \pm\lambda$

Near $q = \lambda$ write $q = \lambda + \varepsilon$. To leading order, $\lambda^2 - q^2 = -2\lambda\varepsilon + O(\varepsilon^2)$. The ODE $-(d/dq)[(\lambda^2 - q^2)(d/dq)\psi] + (2\pi\lambda q)^2\psi = \chi\psi$ becomes, in ε ,

$$(d/d\varepsilon)[2\lambda\varepsilon \cdot (d\psi/d\varepsilon)] + (2\pi\lambda^2)^2 \psi + O(\varepsilon) = \chi \psi.$$

Try $\psi \sim \varepsilon^r$; then $(d/d\varepsilon)[2\lambda\varepsilon \cdot r \cdot \varepsilon^{r-1}] = 2\lambda r^2 \varepsilon^{r-1}$, so the indicial polynomial is $2\lambda r^2 = 0$, giving the double root $r = 0$. The two linearly independent local solutions therefore behave as $\psi_1(\varepsilon) = \text{const} + O(\varepsilon \log \varepsilon)$ and $\psi_2(\varepsilon) = \log|\varepsilon| + O(\varepsilon \log \varepsilon)$. Imposing $(\star)$, $\lim_{q \rightarrow \pm\lambda} (q^2 - \lambda^2) \partial_q \xi = 0$, projects out the ψ_2 component. The boundary trace map (Γ_0, Γ_1) reads off the constant part and the log-coefficient respectively. By symmetry $q \rightarrow -q$ the analysis at $q = -\lambda$ is identical. Together with the L^2 deficiency at $q = \pm\infty$, the deficiency indices of $W_{\{\lambda, \min\}}$ are $(4, 4)$; the self-adjoint extensions form a $U(4)$ family in the Naimark/Krein sense. The Connes–Katsnelson realization is the unique point in $U(4)$ selected by the Fourier-commutation constraint (IMP-T4b).

Appendix B — Identity of Integer Factorization

For any positive integer n with prime factorization $n = \prod_p p^{e_p}$, taking the logarithm gives $\log n = \sum_p e_p \cdot \log p$, an identity in characteristic zero. Applied to $n = q_\chi$ and exponents $e_p(\chi)$ recorded in the conductor factorization, this yields $C_\chi = \log q_\chi$. The identity is purely arithmetic, with no analytic or operator-theoretic content; it cannot be “fit” or “withdrawn”. Its appearance in Tate’s analytic conductor $(q_\chi/\pi)^{\{(s+a_\chi)/2\}}$ as a s -derivative coefficient is what makes the same expression simultaneously the natural Burnol conductor block.

Appendix C — Why the Pure Direct-Sum Closure Is Vacuous in This Sense

For any bounded operator D on a Hilbert space, $\text{Tr}(D^\dagger D) = \|D\|_{\text{HS}}^2 \geq 0$ by definition; equality holds iff $D = 0$. When $D = \oplus_\chi D_\chi$ is block-diagonal, $\text{Tr}(D^\dagger D) = \sum_\chi \text{Tr}(D_\chi^\dagger D_\chi)$, a sum of non-negative reals. Identifying this with a quadratic form $Q(g)$ that depends on a test function g delivers a PSD form by construction. This says nothing about whether Q matches any particular target functional. In the present case the target — the V_4 -decorated Weil form $Q_{\{W, V_4\}}$ — has corpus-PROVEN per-channel indefiniteness (Table 6.1) and corpus-PROVEN non-separable structure (Lemma M31.0). A PSD block-diagonal trace can equal neither: the first because the actual channels are individually signed, the

second because separation is forbidden. The construction is internally consistent as an operator object and PSD by construction; it simply names a different object than the one D4c demands.

References

- [1] A. Connes, “Formules explicites, formules de trace et réalisation spectrale des zéros de la fonction zêta,” Cours au Collège de France (1998), Annu. Collège de France 95, 115–122. [Lemma 6: explicit boundary condition $(q^2 - \lambda^2)\partial_q \xi = 0$.]
- [2] V. Katsnelson, “Self-adjoint boundary conditions for the prolate spheroid differential operator,” in *Indefinite Inner Product Spaces, Schur Analysis, and Differential Equations* (D. Alpay, ed.), Birkhäuser (2016). [Fourier-commuting uniqueness.]
- [3] A. Connes, H. Moscovici, “The UV prolate spectrum matches the zeros of zeta,” *Proc. Natl. Acad. Sci. USA* 119 (22) e2123174119 (2022); arXiv:2112.05500. [$W_{\{\lambda,sa\}}$ spectrum, Sonin compression.]
- [4] J.-P. Ramis, F. Richard-Jung, J. Thomann, “New eigenfunctions for the negative part of the Connes–Moscovici prolate spectrum,” *Comptes Rendus Math.* 363, 1065–1081 (2025); doi:10.5802/crmath.780. [Settling the CCM 2022 conjecture: all non-classical eigenvalues are negative; imaginary-axis characterization.]
- [5] A. Connes, C. Consani, “Weil positivity and trace formula — the archimedean place,” *Selecta Math. (N.S.)* 27, 77 (2021); arXiv:2006.13771.
- [6] A. Connes, C. Consani, H. Moscovici, “Semilocal Sonin space stability,” *Ann. Funct. Anal.* 15:87 (2024); arXiv:2310.18423.
- [7] J. F. Burnol, “Sur les espaces de Sonine associés par de Branges à la transformation de Fourier,” *C. R. Acad. Sci. Paris Ser. I* 335, 689–692 (2002).
- [8] J. F. Burnol, “The explicit formula and the conductor operator,” arXiv:math/9902080 (1999).
- [9] J. Tate, “Fourier analysis in number fields and Hecke’s zeta-functions,” in *Algebraic Number Theory* (J. W. S. Cassels, A. Fröhlich, eds.), Academic Press (1967). [Analytic conductor $(q_\chi/\pi)^{s/2}$.]
- [10] D. Slepian, H. Pollak, “Prolate spheroidal wave functions, Fourier analysis and uncertainty I,” *Bell Syst. Tech. J.* 40, 43–63 (1961). [Concentration operator $A = \Pi_\lambda \Pi_F \Pi_\lambda$.]
- [11] M. Reed, B. Simon, *Methods of Modern Mathematical Physics, Vols. I–II*, Academic Press (1972, 1975). [Spectral theorem, von Neumann–Naimark–Krein self-adjoint extension theory.]
- [12] R. Bhatia, *Matrix Analysis*, Springer GTM 169 (1997). [Inertia, principal angles.]
- [13] Z-Spin Collaboration (K. Kang), ZS-F21 v1.2 — Positive-Square Criterion and Commutator-Coverage Obstruction.
- [14] Z-Spin Collaboration (K. Kang), ZS-M22 v1.0 — ADS-5, §6.4–6.6 (Probe W2 per-channel data; matrix-valued B_K requirement).
- [15] Z-Spin Collaboration (K. Kang), ZS-M23 v1.0 — Dragon D4 (D4a, D4b, D4c, D4d); NC-M23.5, NC-M23.7; O-M23.9, O-M23.11.
- [16] Z-Spin Collaboration (K. Kang), ZS-M28 v1.0 — Locked V_4 channel data; Probe W2 PROVEN diagnostic; Baseline-4 V_4 trace-remainder.
- [17] Z-Spin Collaboration (K. Kang), ZS-M31 v1.0 — Reading C Π_Z -sandwich bilinear form; Lemma M31.0 Non-Separability (18/18); Theorem M31.2 (NEG localization); Theorem M31.4 (Z_2 -parity selection).
- [18] Z-Spin Collaboration (K. Kang), ZS-M33 v1.0 — Path γ -revised colligation; Theorem M33.4 TARGET-SIMULATION.

Version History

v1.0 (March 2026, as ZS-F21 v1.0): closure-type asymmetry; sector–place direction; Mellin frame-duality of the operator-form wall. 18/18 PASS.

v1.1 (March 2026, as ZS-F21 v1.1): positivity wall closed in the obstruction sense — inertia invariance (F21.5), scalar/diagonal no-go (F21.6), minimal matrix-rank obstruction (F21.7), finite semilocal exact inertia (F21.8). 22/22 PASS.

v1.2 (March 2026, as ZS-F21 v1.2): Positive-Square Criterion (F21.9, PROVEN both directions); Commutator-Coverage Obstruction (F21.10, PROVEN negative, null-model 2000 samples); commutator shortcut REJECTED. 26/26 PASS.

v2.0 (March 2026, as ZS-F21 v2.0): Theorem C3.0 (Commutant-Gate Insufficiency, PROVEN this paper, analytic + 200/200 empirical); external imports T4–T5 (Connes 1998 + Katsnelson 2016 + CM 2022 + Ramis–RJ–T 2025 NEW PROVEN); T6d Burnol conductor decoration $C_\chi = \log q_\chi$ DERIVED; T7 pure direct-sum REJECTED via Lemma M31.0; T7’ Π_Z -sandwich V_4 Reading C DERIVED-CONDITIONAL; $\Lambda_\chi = \Lambda\sqrt{(q_\chi)}$ WITHDRAWN; full Weil positivity OPEN / NC. 34/34 PASS, six anti-numerology controls.